Step B:extract characteristics

Cycle	Reference	Discourse	Interpretation	Category	Challenge	Options	Resolution	Concept / Example	Characteristic	Constraints
Cylce 5	54-01	Thoughts are things in the mind.	Thoughts are Things in the mind.	2	Are all thoughts Things in the mind?	Either, (A) all thoughts are Things in the mind, or (B) some thoughts are Things in the mind, or (C) no thoughts are Things in the mind	We chose (B): Thoughts can also be classes (57-03): (the class HUMAN) is not a Thing. Thoughts are also used in other EO theories, with other intensions. We restrict the term "thought" to the triangle and do not consider it a EO term.			
Cylce 6	54-01	Thoughts are things in the mind.	Some thoughts are Things in the mind.				Cycle 5	Thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent concrete things (i.e. things outside the mind), called referents.	6	Are Things outside the mind Concrete things or Referents?	Either (A), the preferred term is Concrete thing, or (B) the preferred term is Referent	We chose (A): Referent, just like thought, is a term of the semiotic triangle and signifies a relation. We do not consider referent to be an EO term.			
	54-02			2	Are Things outside the mind, inside the mind, representations?	Either (A), Thing has subconcepts, or (B), Thing has contradicting characteristics	We chose (A): the mentioned characteristics are assigned to subconcepts later in the theory			
	54-02			2	Aplying EO theory of the previous step leads to a definition loop between Conceptual thing and Concrete thing	Either (A), "represents" is relevant for Concrete thing, or (B) it is relevant for Conceptual thing	We chose (B): it is needed to distinguish Conceptual things from Abstract things			
Cycle 7	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things outside the mind), called referents in the semiotic triangle				Cycle 6	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	outside any mind	
	54-02							Concrete thing	can be represented by a Conceptual thing	cyclic
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.				Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.				Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	3	Are Things always "untouched" ?	Either, (A) a Thing is untouched, or (B) Things are not untouched	We chose (B): mental abstraction does not state a characteristic of a Thing itself. The text refers to Bunge and concflicts with EO statements regarding Conceptual things.			
	55-12			2	Definitional loop between World and Things	Either, (A) a Thing is member of a World, or (B) a World is a collection of Things	We chose (B): a world is a collection of Things. The reverse role will be marked cyclic and not used for definition			
Cycle 8	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things outside the mind), called referents in the semiotic triangle				Cycle 6	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	outside any mind	
	54-02							Concrete thing	can be represented by a Conceptual thing	
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.				Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.				Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.				Cycle 7	World	collection of Things	
	55-12						Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	2	Are Concrete things the same as Things? Is it a world of Concrete things or a world of Things?	Either (A), the premises concern Concrete thing, or (B) the premises concern Things	We chose (B), and allow a model of any collection of Things	World	collection of Things	
Cycle 9	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things outside the mind), called referents in the semiotic triangle				Cycle 6	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	outside any mind	
	54-02							Concrete thing	can be represented by a Conceptual thing	cyclic

	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.			Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.			Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.			Cycle 7	World	collection of Things	
	55-12					Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.			Cycle 8	World	collection of Things	
	56-01						Thing	distinguishable	
	56-01						Thing	can be member of a World	cyclic
	56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.				Thing	with Object, with assigned Form	
	56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.				Thing	with Object, with assigned Form	
	56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.				[a cloud]	with Object, with assigned Form, Thing	
	56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.				Thing	with Object, with assigned Form	
	56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.				[a body]	with Object, with assigned Form	
	56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.				Thing	with Object, with assigned Form	
	56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.				Thing	with Object	cyclic
	56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.				Thing	with Object, with assigned Form	
	56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.			Object formal context	Concrete thing	with Concrete object	cyclic
	56-19						[a human]	Concrete thing, with Concrete object	
	56-19						[a dog]	Concrete thing, with Concrete object	
	56-19						[a membership]	Concrete thing, with Concrete object	
	56-19						[a rental]	Concrete thing, with Concrete object	
	56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.				Abstract thing	with Abstract object	cyclic
	56-20						<a number>	Abstract thing, with Abstract object	
	56-20						<a boolean value>	Abstract thing, with Abstract object	
	56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.			Cycle 8	World	collection of Things	
	56-21						Conceptual thing	can represent a Concrete thing	
	56-21						Concrete thing	can be represented by a Conceptual thing	cyclic
	56-21					Object formal context	Concrete thing	with Concrete object	cyclic
	56-21					Object formal context	Conceptual thing	with Conceptual object	cyclic
	56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.	2	Definitional loop due to the relationship between Class en Thing	Either, (A) the relationship is defined with Thing, or (B) the relationship is defined with Class	We chose (B), a Class is a collection of Things		
Cycle 10	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.			Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things outside the mind), called referents in the semiotic triangle			Cycle 6	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02						Concrete thing	outside any mind	
	54-02						Concrete thing	can be represented by a Conceptual thing	cyclic

54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.		Cycle 5		
54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.		Cycle 5		
55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.		Cycle 7	World	collection of Things
55-12				Cycle 7	Thing	can be member of a World	cyclic
56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.		Cycle 8	World	collection of Things
56-01					Thing	distinguishable
56-01					Thing	can be member of a World	cyclic
56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.			Thing	with Object, with assigned Form
56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.			Thing	with Object, with assigned Form
56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.			[a cloud]	with Object, with assigned Form, Thing
56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.			Thing	with Object, with assigned Form
56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.			[a body]	with Object, with assigned Form
56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.			Thing	with Object, with assigned Form
56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.			Thing	with Object	cyclic
56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.			Thing	with Object, with assigned Form
56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.		Object formal context	Concrete thing	with Concrete object	cyclic
56-19					[a human]	Concrete thing, with Concrete object
56-19					[a dog]	Concrete thing, with Concrete object
56-19					[a membership]	Concrete thing, with Concrete object
56-19					[a rental]	Concrete thing, with Concrete object
56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.			Abstract thing	with Abstract object	cyclic
56-20					<a number>	Abstract thing, with Abstract object
56-20					<a boolean value>	Abstract thing, with Abstract object
56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.		Cycle 8	World	collection of Things
56-21					Conceptual thing	can represent a Concrete thing
56-21					Concrete thing	can be represented by a Conceptual thing	cyclic
56-21				Object formal context	Concrete thing	with Concrete object	cyclic
56-21				Object formal context	Conceptual thing	with Conceptual object	cyclic
56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.		Cycle 9	Class	collection of Things
56-24				Cycle 9	Thing	can be member of a Class	cyclic
56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.		Cycle 9	Class	collection of Things
56-27				Cycle 9	Thing	can be member of a Class	cyclic
56-27			-	We use [HUMAN] instead of [humans], in correspondence with 57-02	[HUMAN]	Class, collection of Things
56-27			-	We use [RENTAL], based on 57-02	[RENTAL]	Class, collection of Things

	56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.				Conceptual thing	with Conceptual object	cyclic
	56-28						Conceptual thing	with Object, with assigned Form	
	56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.	2	Is observable the characteristic that distinguishes tangible and intangible? In Object formal context we concluded that observable does not fit that role.	Either, (A) observable distinguish (in)tangible, or (B) another characteristic distinguishes the subconcepts	We chose (B); the glossary states that Tangible things have physical dimensions while Intangible things do not.		
	56-29			2	Definitional loop between Conceived world and Intangible concrete things	Either, (A) an Intangible thing is member of a Conceived world, or (B) a Conceived world is a collection of Intangible things	We chose (B); the glossary states that Tangible things have physical dimensions while Intangible things do not.		
Cycle 11	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things outside the mind), called referents in the semiotic triangle				Cycle 6	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	outside any mind	
	54-02							Concrete thing	can be represented by a Conceptual thing	cyclic
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.				Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.				Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.				Cycle 7	World	collection of Things	
	55-12						Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.				Cycle 8	World	collection of Things	
	56-01							Thing	distinguishable	
	56-01							Thing	can be member of a World	cyclic
	56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.					Thing	with Object, with assigned Form	
	56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.					Thing	with Object, with assigned Form	
	56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.					[a cloud]	with Object, with assigned Form, Thing	
	56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.					Thing	with Object, with assigned Form	
	56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.					[a body]	with Object, with assigned Form	
	56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.					Thing	with Object, with assigned Form	
	56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.					Thing	with Object	cyclic
	56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.					Thing	with Object, with assigned Form	
	56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.				Object formal context	Concrete thing	with Concrete object	cyclic
	56-19							[a human]	Concrete thing, with Concrete object	
	56-19							[a dog]	Concrete thing, with Concrete object	
	56-19							[a membership]	Concrete thing, with Concrete object	
	56-19							[a rental]	Concrete thing, with Concrete object	
	56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.					Abstract thing	with Abstract object	cyclic
	56-20							<a number>	Abstract thing, with Abstract object	
	56-20							<a boolean value>	Abstract thing, with Abstract object	

	56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.				Cycle 8	World	collection of Things	
	56-21							Conceptual thing	can represent a Concrete thing	
	56-21							Concrete thing	can be represented by a Conceptual thing	cyclic
	56-21						Object formal context	Concrete thing	with Concrete object	cyclic
	56-21						Object formal context	Conceptual thing	with Conceptual object	cyclic
	56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.				Cycle 9	Class	collection of Things	
	56-24						Cycle 9	Thing	can be member of a Class	cyclic
	56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.				Cycle 9	Class	collection of Things	
	56-27						Cycle 9	Thing	can be member of a Class	cyclic
	56-27							[HUMAN]	Class, collection of Things	
	56-27							[RENTAL]	Class, collection of Things	
	56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.					Conceptual thing	with Conceptual object	cyclic
	56-28							Conceptual thing	with Object, with assigned Form	
	56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.				Cycle 10	Tangible concrete thing	physical dimensions	disjoint & complete: Tangible concrete thing, Intangible concrete thing
	56-29							Intangible concrete thing	conceived	
	56-29							Intangible concrete thing	can be member of a conceived world	cyclic
	56-29							Conceived world	collection of Intangible things	
	56-29							[a human]	Concrete thing, Tangible concrete thing, physical dimensions	
	56-29							[a dog]	Concrete thing, Tangible concrete thing, physical dimensions	
	56-29							[a membership]	Concrete thing, Intangible concrete thing, conceived	
	56-29							[a rental]	Concrete thing, Intangible concrete thing, conceived	
	56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.					Concrete thing	instance of a Type	
	56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).							
	56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	-	The bracket convention is not consistent		We follow the convention esthablished in the Object formal context	[the human 'Charlie Chaplin']	Concrete thing	
	56-33							(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
	56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and Classes, we use the bracket pair "[" and "]".							
	57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to Classes.							
	57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.					[HUMAN]	Concrete class, collection of Concrete things	
	57-02							Concrete class	Collection of Concrete things	
	57-03	It is represented in our mind by the thought (the class HUMAN).	It is represented in our mind by the thought (the class HUMAN).	2	A class representation of another class is second order logic, (statement on sets), while EO states to be well defined in first order logic	Either, (A) EO is not well defined in first order logic, or (B) the logical statement can be rephrased in first-order logic	We chose (B); the Conceptual things in the Conceptual class represent Concrete things in the Concrete class; the relationship is indirect.			
Cycle 12	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things outside the mind), called referents in the semiotic triangle				Cycle 6	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	outside any mind	
	54-02							Concrete thing	can be represented by a Conceptual thing	cyclic

54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Cycle 5			
54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	Cycle 5			
55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.	Cycle 7	World	collection of Things	
55-12			Cycle 7	Thing	can be member of a World	cyclic
56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.	Cycle 8	World	collection of Things	
56-01				Thing	distinguishable	
56-01				Thing	can be member of a World	cyclic
56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.		Thing	with Object, with assigned Form	
56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.		Thing	with Object, with assigned Form	
56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.		[a cloud]	with Object, with assigned Form, Thing	
56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.		Thing	with Object, with assigned Form	
56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.		[a body]	with Object, with assigned Form	
56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.		Thing	with Object, with assigned Form	
56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.		Thing	with Object	cyclic
56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.		Thing	with Object, with assigned Form	
56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Object formal context	Concrete thing	with Concrete object	cyclic
56-19				[a human]	Concrete thing, with Concrete object	
56-19				[a dog]	Concrete thing, with Concrete object	
56-19				[a membership]	Concrete thing, with Concrete object	
56-19				[a rental]	Concrete thing, with Concrete object	
56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.		Abstract thing	with Abstract object	cyclic
56-20				<a number>	Abstract thing, with Abstract object	
56-20				<a boolean value>	Abstract thing, with Abstract object	
56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	Cycle 8	World	collection of Things	
56-21				Conceptual thing	can represent a Concrete thing	
56-21				Concrete thing	can be represented by a Conceptual thing	cyclic
56-21			Object formal context	Concrete thing	with Concrete object	cyclic
56-21			Object formal context	Conceptual thing	with Conceptual object	cyclic
56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.	Cycle 9	Class	collection of Things	
56-24			Cycle 9	Thing	can be member of a Class	cyclic
56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.	Cycle 9	Class	collection of Things	
56-27			Cycle 9	Thing	can be member of a Class	cyclic
56-27				[HUMAN]	Class, collection of Things	
56-27				[RENTAL]	Class, collection of Things	

56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.					Conceptual thing	with Conceptual object	cyclic
56-28							Conceptual thing	with Object, with assigned Form	
56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.				Cycle 10	Tangible concrete thing	physical dimensions	disjoint & complete: Tangible concrete thing, Intangible concrete thing
							Intangible concrete thing	conceived	
56-29							Intangible concrete thing	can be member of a conceived world	cyclic
56-29							Conceived world	collection of Intangible things	
56-29							[a human]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29							[a dog]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29							[a membership]	Concrete thing, Intangible concrete thing, conceived	
56-29							[a rental]	Concrete thing, Intangible concrete thing, conceived	
56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.					Concrete thing	instance of a Type	
56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).							
56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	-	The bracket convention is not consistent		We follow the convention esthablished in the Object formal context	[the human 'Charlie Chaplin']	Concrete thing	
56-33							(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and Classes, we use the bracket pair "[" and "]".							
57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to Classes.							
57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.					[HUMAN]	Concrete class, collection of Concrete things	
57-02							Concrete class	Collection of Concrete things	
57-03	It is represented in our mind by the thought (the class HUMAN).	The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN)					(HUMAN)	Conceptual class, collectiong of Conceptual things	
57-04	The string "HUMAN" is a name from the name class for classes.	The string "HUMAN" is a name from the name class for classes.							
57-06	The referents we consider now are thus the Concrete objects of Concrete things.	The referents we consider now are thus the Concrete objects of Concrete things.							
57-09	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.							
58-02	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.					Abstract thing	with Abstract object	cyclic
58-03	As indicated by the question mark, Abstract things don't represent any Concrete things	As indicated by the question mark, Abstract things don't represent any Concrete things					Abstract thing	can not represent a Concrete thing	cyclic
58-04	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.					Concrete thing	with Concrete object	cyclic
58-04							Abstract thing	with Abstract object	cyclic
58-04									disjoint: Concrete thing, Abstract thing
58-05	Well-known abstract things are numbers.	Well-known abstract things are numbers.					<a number>	Abstract thing	
58-06	The name "the number '387'" signifies the abstract thing (the number '387').	The name "the number '387'" signifies the abstract thing (the number '387').					<a number>	Abstract thing	
58-07	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.					<a number>	Abstract thing	
58-08	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.					Thing	distinguishable	
58-14	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.					(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
59-01	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	6	Entity is a synonym of Concrete thing?	Either (A), Entity is the preferred term, or (B), Concrete thing is the preferred term, or (C),they are distinct concepts	We chose (C):they are distinct concepts. The next sentence states that Entity refers to an Concrete unary fact. Entity is a synoym of that term.			

Cycle 13	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.	Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things outside the mind), called referents in the semiotic triangle	Cycle 6	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02				Concrete thing	outside any mind	
	54-02				Concrete thing	can be represented by a Conceptual thing	cyclic
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.	Cycle 7	World	collection of Things	
	55-12			Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.	Cycle 8	World	collection of Things	
	56-01				Thing	distinguishable	
	56-01				Thing	can be member of a World	cyclic
	56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.		Thing	with Object, with assigned Form	
	56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.		Thing	with Object, with assigned Form	
	56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.		[a cloud]	with Object, with assigned Form, Thing	
	56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.		Thing	with Object, with assigned Form	
	56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.		[a body]	with Object, with assigned Form	
	56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.		Thing	with Object, with assigned Form	
	56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.		Thing	with Object	cyclic
	56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.		Thing	with Object, with assigned Form	
	56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Object formal context	Concrete thing	with Concrete object	cyclic
	56-19				[a human]	Concrete thing, with Concrete object	
	56-19				[a dog]	Concrete thing, with Concrete object	
	56-19				[a membership]	Concrete thing, with Concrete object	
	56-19				[a rental]	Concrete thing, with Concrete object	
	56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.		Abstract thing	with Abstract object	cyclic
	56-20				<a number>	Abstract thing, with Abstract object	
	56-20				<a boolean value>	Abstract thing, with Abstract object	
	56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	Cycle 8	World	collection of Things	
	56-21				Conceptual thing	can represent a Concrete thing	
	56-21				Concrete thing	can be represented by a Conceptual thing	cyclic
	56-21			Object formal context	Concrete thing	with Concrete object	cyclic
	56-21			Object formal context	Conceptual thing	with Conceptual object	cyclic
	56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.	Cycle 9	Class	collection of Things	
	56-24			Cycle 9	Thing	can be member of a Class	cyclic

56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.			Cycle 9	Class	collection of Things	
56-27					Cycle 9	Thing	can be member of a Class	cyclic
56-27						[HUMAN]	Class, collection of Things	
56-27						[RENTAL]	Class, collection of Things	
56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.				Conceptual thing	with Conceptual object	cyclic
56-28						Conceptual thing	with Object, with assigned Form	
56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.			Cycle 10	Tangible concrete thing	physical dimensions	disjoint & complete: Tangible concrete thing, Intangible concrete thing
						Intangible concrete thing	conceived	
56-29						Intangible concrete thing	can be member of a conceived world	cyclic
56-29						Conceived world	collection of Intangible things	
56-29						[a human]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a dog]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a membership]	Concrete thing, Intangible concrete thing, conceived	
56-29						[a rental]	Concrete thing, Intangible concrete thing, conceived	
56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.				Concrete thing	instance of a Type	
56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).						
56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	-	The bracket convention is not consistent	We follow the convention esthablished in the Object formal context	[the human 'Charlie Chaplin']	Concrete thing	
56-33						(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and Classes, we use the bracket pair "[" and "]".						
57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to Classes.						
57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.				[HUMAN]	Concrete class, collection of Concrete things	
57-02						Concrete class	Collection of Concrete things	
57-03	It is represented in our mind by the thought (the class HUMAN).	The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN)				(HUMAN)	Conceptual class, collectiong of Conceptual things	
57-04	The string "HUMAN" is a name from the name class for classes.	The string "HUMAN" is a name from the name class for classes.						
57-06	The referents we consider now are thus the Concrete objects of Concrete things.	The referents we consider now are thus the Concrete objects of Concrete things.						
57-09	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.						
58-02	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.				Abstract thing	with Abstract object	cyclic
58-03	As indicated by the question mark, Abstract things don't represent any Concrete things	As indicated by the question mark, Abstract things don't represent any Concrete things				Abstract thing	can not represent a Concrete thing	cyclic
58-04	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.				Concrete thing	with Concrete object	cyclic
58-04						Abstract thing	with Abstract object	cyclic
58-04								disjoint: Concrete thing, Abstract thing
58-05	Well-known abstract things are numbers.	Well-known abstract things are numbers.				<a number>	Abstract thing	
58-06	The name "the number '387'" signifies the abstract thing (the number '387').	The name "the number '387'" signifies the abstract thing (the number '387').				<a number>	Abstract thing	
58-07	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.				<a number>	Abstract thing	
58-08	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.				Thing	distinguishable	

	58-14	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.				(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
	59-01	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	If the Thing of a Unary fact is a Concrete thing, the Concrete unary fact is commonly called an Entity.			cycle 12	Thing	variable of Unary fact	cyclic
	59-01						Concrete thing	variable of Entity	cyclic
	59-02	As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".	As an example, one commonly speaks of the Entity Charlie Chaplin, thereby referring to the Concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".				[a human]	Concrete thing	
	59-03	Likewise, if the thing of a unary fact is Abstract, the Abstract thing is commonly called a Value.	Likewise, if the Thing of a Unary fact is Abstract, the Abstract unary fact is commonly called a Value.			cycle 12	Thing	variable of Unary fact	cyclic
	59-03						Abstract thing	variable of Value	cyclic
	59-04	As an example, day 2411108 is a value on the time scale Julian Date.	As an example, day 2411108 is a value on the time scale Julian Date.	3 is a day an Abstract thing or a Concrete thing?	Either (A), day 2411108 is a Value, or (B), day 2411108 is an Entity	We chose (B): the example is an Entity instead of a Value. An Abstract thing does not have a referent, but a day does have a referent. However, the date 2411108 is a Value, on the time scale Julian Date. The day itself is not.			
Cycle 14	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.			Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things outside the mind), called referents in the semiotic triangle			Cycle 6	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02						Concrete thing	outside any mind	
	54-02						Concrete thing	can be represented by a Conceptual thing	cyclic
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.			Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.			Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.			Cycle 7	World	collection of Things	
	55-12					Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.			Cycle 8	World	collection of Things	
	56-01						Thing	distinguishable	
	56-01						Thing	can be member of a World	cyclic
	56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.				Thing	with Object, with assigned Form	
	56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.				Thing	with Object, with assigned Form	
	56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.				[a cloud]	with Object, with assigned Form, Thing	
	56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.				Thing	with Object, with assigned Form	
	56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.				[a body]	with Object, with assigned Form	
	56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.				Thing	with Object, with assigned Form	
	56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.				Thing	with Object	cyclic
	56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.				Thing	with Object, with assigned Form	
	56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.			Object formal context	Concrete thing	with Concrete object	cyclic
	56-19						[a human]	Concrete thing, with Concrete object	
	56-19						[a dog]	Concrete thing, with Concrete object	
	56-19						[a membership]	Concrete thing, with Concrete object	
	56-19						[a rental]	Concrete thing, with Concrete object	

56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.				Abstract thing	with Abstract object	cyclic
56-20						<a number>	Abstract thing, with Abstract object	
56-20						<a boolean value>	Abstract thing, with Abstract object	
56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.			Cycle 8	World	collection of Things	
56-21						Conceptual thing	can represent a Concrete thing	
56-21						Concrete thing	can be represented by a Conceptual thing	cyclic
56-21					Object formal context	Concrete thing	with Concrete object	cyclic
56-21					Object formal context	Conceptual thing	with Conceptual object	cyclic
56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.			Cycle 9	Class	collection of Things	
56-24					Cycle 9	Thing	can be member of a Class	cyclic
56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.			Cycle 9	Class	collection of Things	
56-27					Cycle 9	Thing	can be member of a Class	cyclic
56-27						[HUMAN]	Class, collection of Things	
56-27						[RENTAL]	Class, collection of Things	
56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.				Conceptual thing	with Conceptual object	cyclic
56-28						Conceptual thing	with Object, with assigned Form	
56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.			Cycle 10	Tangible concrete thing	physical dimensions	disjoint & complete: Tangible concrete thing, Intangible concrete thing
56-29						Intangible concrete thing	conceived	
56-29						Intangible concrete thing	can be member of a conceived world	cyclic
56-29						Conceived world	collection of Intangible things	
56-29						[a human]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a dog]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a membership]	Concrete thing, Intangible concrete thing, conceived	
56-29						[a rental]	Concrete thing, Intangible concrete thing, conceived	
56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.				Concrete thing	instance of a Type	
56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).						
56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	-	The bracket convention is not consistent	We follow the convention esthablished in the Object formal context	[the human 'Charlie Chaplin']	Concrete thing	
56-33						(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and Classes, we use the bracket pair "[" and "]".						
57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to Classes.						
57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.				[HUMAN]	Concrete class, collection of Concrete things	
57-02						Concrete class	Collection of Concrete things	
57-03	It is represented in our mind by the thought (the class HUMAN).	The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN)				(HUMAN)	Conceptual class, collectiong of Conceptual things	
57-04	The string "HUMAN" is a name from the name class for classes.	The string "HUMAN" is a name from the name class for classes.						
57-06	The referents we consider now are thus the Concrete objects of Concrete things.	The referents we consider now are thus the Concrete objects of Concrete things.						
57-09	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.						

	58-02	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.					Abstract thing	with Abstract object	cyclic
	58-03	As indicated by the question mark, Abstract things don't represent any Concrete things	As indicated by the question mark, Abstract things don't represent any Concrete things					Abstract thing	can not represent a Concrete thing	cyclic
	58-04	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.					Concrete thing	with Concrete object	cyclic
	58-04							Abstract thing	with Abstract object	cyclic
	58-04									disjoint: Concrete thing, Abstract thing
	58-05	Well-known abstract things are numbers.	Well-known abstract things are numbers.					<a number>	Abstract thing	
	58-06	The name "the number '387'" signifies the abstract thing (the number '387').	The name "the number '387'" signifies the abstract thing (the number '387').					<a number>	Abstract thing	
	58-07	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.					<a number>	Abstract thing	
	58-08	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.					Thing	distinguishable	
	58-14	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.					(the human 'Charlie Chaplin')	Conceptual thing	
	59-01	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	If the Thing of a Unary fact is a Concrete thing, the Concrete unary fact is commonly called an Entity.				cycle 12	Thing	variable of Unary fact	cyclic
	59-01							Concrete thing	variable of Entity, Thing	cyclic
	59-02	As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".	As an example, one commonly speaks of the Entity Charlie Chaplin, thereby referring to the Concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".					[a human]	Concrete thing	
	59-03	Likewise, if the thing of a unary fact is Abstract, the Abstract thing is commonly called a Value.	Likewise, if the Thing of a Unary fact is Abstract, the Abstract unary fact is commonly called a Value.				cycle 12	Thing	variable of Unary fact	cyclic
	59-03							Abstract thing	variable of Value, Thing	cyclic
	59-04	As an example, day 2411108 is a value on the time scale Julian Date.	As an example, date 2411108 is a value on the time scale Julian Date.				cycle 13	<a date>	Abstract thing, Thing	
	59-07	In answering these questions, we will adhere strictly to the distinctions we have made between concrete and conceptual things.	In answering these questions, we will adhere strictly to the distinctions we have made between Concrete and Conceptual things.					Concrete thing		disjoint: Concrete thing, Conceptual thing
	59-07							Conceptual thing		
	59-08	Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.	Concrete things are the Things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.	6	"outside any mind" (54-02) and "in reality" (59-08) signify the same characteristic.	Either (A), "outside any mind", or (B), "in reality" is the preferred term	We chose (B). Both are only used once, although "in reality" is also used with Concrete fact. A positive characteristic is preferred instead of a negative one, and "outside any mind" is with semiotic theory of referents instead of defining Concrete thing.	Concrete thing	Thing, in reality	disjoint & complete: inside a mind, in reality
Cycle 15	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things in reality), called referents in the semiotic triangle				Cycle 6, 14	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	in reality	
	54-02							Concrete thing	can be represented by a Conceptual thing	cyclic
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.				Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.				Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.				Cycle 7	World	collection of Things	
	55-12						Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.				Cycle 8	World	collection of Things	
	56-01							Thing	distinguishable	
	56-01							Thing	can be member of a World	cyclic
	56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.					Thing	with Object, with assigned Form	

56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.		Thing	with Object, with assigned Form	
56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.		[a cloud]	with Object, with assigned Form, Thing	
56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.		Thing	with Object, with assigned Form	
56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.		[a body]	with Object, with assigned Form	
56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.		Thing	with Object, with assigned Form	
56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.		Thing	with Object	cyclic
56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.		Thing	with Object, with assigned Form	
56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Object formal context	Concrete thing	with Concrete object	cyclic
56-19				[a human]	Concrete thing, with Concrete object	
56-19				[a dog]	Concrete thing, with Concrete object	
56-19				[a membership]	Concrete thing, with Concrete object	
56-19				[a rental]	Concrete thing, with Concrete object	
56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.		Abstract thing	with Abstract object	cyclic
56-20				<a number>	Abstract thing, with Abstract object	
56-20				<a boolean value>	Abstract thing, with Abstract object	
56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	Cycle 8	World	collection of Things	
56-21				Conceptual thing	can represent a Concrete thing	
56-21				Concrete thing	can be represented by a Conceptual thing	cyclic
56-21			Object formal context	Concrete thing	with Concrete object	cyclic
56-21			Object formal context	Conceptual thing	with Conceptual object	cyclic
56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.	Cycle 9	Class	collection of Things	
56-24			Cycle 9	Thing	can be member of a Class	cyclic
56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.	Cycle 9	Class	collection of Things	
56-27			Cycle 9	Thing	can be member of a Class	cyclic
56-27				[HUMAN]	Class, collection of Things	
56-27				[RENTAL]	Class, collection of Things	
56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.		Conceptual thing	with Conceptual object	cyclic
56-28				Conceptual thing	with Object, with assigned Form	
56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.	Cycle 10	Tangible concrete thing	physical dimensions	disjoint & complete: Tangible concrete thing, Intangible concrete thing
				Intangible concrete thing	conceived	
56-29				Intangible concrete thing	can be member of a conceived world	cyclic
56-29				Conceived world	collection of Intangible things	
56-29				[a human]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29				[a dog]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29				[a membership]	Concrete thing, Intangible concrete thing, conceived	
56-29				[a rental]	Concrete thing, Intangible concrete thing, conceived	

56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.				Concrete thing	instance of a Type	
56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).						
56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	-	The bracket convention is not consistent	We follow the convention esthablished in the Object formal context	[the human 'Charlie Chaplin']	Concrete thing	
56-33						(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and Classes, we use the bracket pair "[" and "]".						
57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to Classes.						
57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.				[HUMAN]	Concrete class, collection of Concrete things	
57-02						Concrete class	Collection of Concrete things	
57-03	It is represented in our mind by the thought (the class HUMAN).	The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN)				(HUMAN)	Conceptual class, collectiong of Conceptual things	
57-04	The string "HUMAN" is a name from the name class for classes.	The string "HUMAN" is a name from the name class for classes.						
57-06	The referents we consider now are thus the Concrete objects of Concrete things.	The referents we consider now are thus the Concrete objects of Concrete things.						
57-09	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.						
58-02	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.				Abstract thing	with Abstract object	cyclic
58-03	As indicated by the question mark, Abstract things don't represent any Concrete things	As indicated by the question mark, Abstract things don't represent any Concrete things				Abstract thing	can not represent a Concrete thing	cyclic
58-04	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.				Concrete thing	with Concrete object	cyclic
58-04						Abstract thing	with Abstract object	cyclic
58-04								disjoint: Concrete thing, Abstract thing
58-05	Well-known abstract things are numbers.	Well-known abstract things are numbers.				<a number>	Abstract thing	
58-06	The name "the number '387'" signifies the abstract thing (the number '387').	The name "the number '387'" signifies the abstract thing (the number '387').				<a number>	Abstract thing	
58-07	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.				<a number>	Abstract thing	
58-08	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.				Thing	distinguishable	
58-14	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.				(the human 'Charlie Chaplin')	Conceptual thing	
59-01	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	If the Thing of a Unary fact is a Concrete thing, the Concrete unary fact is commonly called an Entity.			cycle 12	Thing	variable of Unary fact	cyclic
59-01						Concrete thing	variable of Entity, Thing	cyclic
59-02	As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".	As an example, one commonly speaks of the Entity Charlie Chaplin, thereby referring to the Concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".				[a human]	Concrete thing	
59-03	Likewise, if the thing of a unary fact is Abstract, the Abstract thing is commonly called a Value.	Likewise, if the Thing of a Unary fact is Abstract, the Abstract unary fact is commonly called a Value.			cycle 12	Thing	variable of Unary fact	cyclic
59-03						Abstract thing	variable of Value, Thing	cyclic
59-04	As an example, day 2411108 is a value on the time scale Julian Date.	As an example, date 2411108 is a value on the time scale Julian Date.			cycle 13	<a date>	Abstract thing, Thing	
59-07	In answering these questions, we will adhere strictly to the distinctions we have made between concrete and conceptual things.	In answering these questions, we will adhere strictly to the distinctions we have made between Concrete and Conceptual things.				Concrete thing		disjoint: Concrete thing, Conceptual thing
59-07						Conceptual thing		
59-08	Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.	Concrete things are the Things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.			Cycle 15	Concrete thing	Thing, in reality	disjoint & complete: inside a mind, in reality
59-08						[an insurance policy]	Concrete thing, in reality, Thing	

	59-08							[a human]	Concrete thing, in reality, Thing	
	59-08							[a car]	Concrete thing, in reality, Thing	
	59-08							[a house]	Concrete thing, in reality, Thing	
	59-08							[a rental]	Concrete thing, in reality, Thing	
	59-08							[an employment contract]	Concrete thing, in reality, Thing	
	59-09	As said, concrete things can be tangible (like a house) and intangible (like a rental).	As said, concrete things can be tangible (like a house) and intangible (like a rental).					Tangible concrete thing	Concrete thing	disjoint: Tangible concrete thing, Intangible concrete thing
	59-09							Intangible concrete thing	Concrete thing	
	59-09							[a house]	Concrete thing, Tangible concrete thing	
	59-09							[a rental]	Concrete thing, Intangible concrete thing	
	59-10	Conceptual things are things in the mind.	Conceptual things are Things in the mind.					Conceptual thing	Thing, in the mind	
	60-05	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).					Concrete thing		
	60-05							Tangible concrete thing		
	60-05							Intangible concrete thing		
	60-06	The Form of a Thing is the collection of its Features.	The Form of a Concrete thing is the collection of its Features.					Concrete thing	with assigned Form	
	60-08	In applying a type, the mind puts, so to speak, the type's template over the things in the world, and sieves out the ones that conform.	In applying a type, the mind puts, so to speak, the type's template over the Things in the World, and sieves out the ones that conform.	6	does a Thing conform to a Type or is it an instance of a Type (56-30)? Both statements have the premise that the Form of a Thing is a superset of a Type intension	Either, (A) a thing conforms to a Type, or (B) a Thing is an instance of a Type	We chose (A). Due to the sameness and change solution, it is -not- possible to create an instance of an Type, which instance suggests.			
Cycle 16	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things in reality), called referents in the semiotic triangle				Cycle 6, 14	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	in reality	
	54-02							Concrete thing	can be represented by a Conceptual thing	cyclic
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.				Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.				Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.				Cycle 7	World	collection of Things	
	55-12						Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.				Cycle 8	World	collection of Things	
	56-01							Thing	distinguishable	
	56-01							Thing	can be member of a World	cyclic
	56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.					Thing	with Object, with assigned Form	
	56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.					Thing	with Object, with assigned Form	
	56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.					[a cloud]	with Object, with assigned Form, Thing	
	56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.					Thing	with Object, with assigned Form	
	56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.					[a body]	with Object, with assigned Form	
	56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object.					Thing	with Object, with assigned Form	
		Bunge names an Object a bare individual.	Bunge names an Object a bare individual.							
	56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.					Thing	with Object	cyclic
	56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.					Thing	with Object, with assigned Form	
	56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.				Object formal context	Concrete thing	with Concrete object	cyclic
	56-19							[a human]	Concrete thing, with Concrete object	
	56-19							[a dog]	Concrete thing, with Concrete object	
	56-19							[a membership]	Concrete thing, with Concrete object	
	56-19							[a rental]	Concrete thing, with Concrete object	

56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.				Abstract thing	with Abstract object	cyclic
56-20						<a number>	Abstract thing, with Abstract object	
56-20						<a boolean value>	Abstract thing, with Abstract object	
56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.			Cycle 8	World	collection of Things	
56-21						Conceptual thing	can represent a Concrete thing	
56-21						Concrete thing	can be represented by a Conceptual thing	cyclic
56-21					Object formal context	Concrete thing	with Concrete object	cyclic
56-21					Object formal context	Conceptual thing	with Conceptual object	cyclic
56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.			Cycle 9	Class	collection of Things	
56-24					Cycle 9	Thing	can be member of a Class	cyclic
56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.			Cycle 9	Class	collection of Things	
56-27					Cycle 9	Thing	can be member of a Class	cyclic
56-27						[HUMAN]	Class, collection of Things	
56-27						[RENTAL]	Class, collection of Things	
56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.				Conceptual thing	with Conceptual object	cyclic
56-28						Conceptual thing	with Object, with assigned Form	
56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.			Cycle 10	Tangible concrete thing	physical dimensions	disjoint & complete: Tangible concrete thing, Intangible concrete thing
56-29						Intangible concrete thing	conceived	
56-29						Intangible concrete thing	can be member of a conceived world	cyclic
56-29						Conceived world	collection of Intangible things	
56-29						[a human]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a dog]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a membership]	Concrete thing, Intangible concrete thing, conceived	
56-29						[a rental]	Concrete thing, Intangible concrete thing, conceived	
56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.				Concrete thing	can conform to a Type	
56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).						
56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	-	The bracket convention is not consistent	We follow the convention esthablished in the Object formal context	[the human 'Charlie Chaplin']	Concrete thing	
56-33						(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and Classes, we use the bracket pair "[" and "]".						
57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to Classes.						
57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.				[HUMAN]	Concrete class, collection of Concrete things	
57-02						Concrete class	Collection of Concrete things	
57-03	It is represented in our mind by the thought (the class HUMAN).	The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN)				(HUMAN)	Conceptual class, collectiong of Conceptual things	
57-04	The string "HUMAN" is a name from the name class for classes.	The string "HUMAN" is a name from the name class for classes.						
57-06	The referents we consider now are thus the Concrete objects of Concrete things.	The referents we consider now are thus the Concrete objects of Concrete things.						
57-09	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.						

58-02	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.				Abstract thing	with Abstract object	cyclic
58-03	As indicated by the question mark, Abstract things don't represent any Concrete things	As indicated by the question mark, Abstract things don't represent any Concrete things				Abstract thing	can not represent a Concrete thing	cyclic
58-04	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.				Concrete thing	with Concrete object	cyclic
58-04						Abstract thing	with Abstract object	cyclic
58-04								disjoint: Concrete thing, Abstract thing
58-05	Well-known abstract things are numbers.	Well-known abstract things are numbers.				<a number>	Abstract thing	
58-06	The name "the number '387'" signifies the abstract thing (the number '387').	The name "the number '387'" signifies the abstract thing (the number '387').				<a number>	Abstract thing	
58-07	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.				<a number>	Abstract thing	
58-08	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.				Thing	distinguishable	
58-14	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.				(the human 'Charlie Chaplin')	Conceptual thing	
59-01	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	If the Thing of a Unary fact is a Concrete thing, the Concrete unary fact is commonly called an Entity.			cycle 12	Thing	variable of Unary fact	cyclic
59-01						Concrete thing	variable of Entity, Thing	cyclic
59-02	As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".	As an example, one commonly speaks of the Entity Charlie Chaplin, thereby referring to the Concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".				[a human]	Concrete thing	
59-03	Likewise, if the thing of a unary fact is Abstract, the Abstract thing is commonly called a Value.	Likewise, if the Thing of a Unary fact is Abstract, the Abstract unary fact is commonly called a Value.			cycle 12	Thing	variable of Unary fact	cyclic
59-03						Abstract thing	variable of Value, Thing	cyclic
59-04	As an example, day 2411108 is a value on the time scale Julian Date.	As an example, date 2411108 is a value on the time scale Julian Date.			cycle 13	<a date>	Abstract thing, Thing	
59-07	In answering these questions, we will adhere strictly to the distinctions we have made between concrete and conceptual things.	In answering these questions, we will adhere strictly to the distinctions we have made between Concrete and Conceptual things.				Concrete thing		disjoint: Concrete thing, Conceptual thing
59-07						Conceptual thing		
59-08	Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.	Concrete things are the Things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.			Cycle 15	Concrete thing	Thing, in reality	disjoint & complete: inside a mind, in reality
59-08						[an insurance policy]	Concrete thing, in reality, Thing	
59-08						[a human]	Concrete thing, in reality, Thing	
59-08						[a car]	Concrete thing, in reality, Thing	
59-08						[a house]	Concrete thing, in reality, Thing	
59-08						[a rental]	Concrete thing, in reality, Thing	
59-08						[an employment contract]	Concrete thing, in reality, Thing	
59-09	As said, concrete things can be tangible (like a house) and intangible (like a rental).	As said, concrete things can be tangible (like a house) and intangible (like a rental).				Tangible concrete thing	Concrete thing	disjoint: Tangible concrete thing, Intangible concrete thing
59-09						Intangible concrete thing	Concrete thing	
59-09						[a house]	Concrete thing, Tangible concrete thing	
59-09						[a rental]	Concrete thing, Intangible concrete thing	
59-10	Conceptual things are things in the mind.	Conceptual things are Things in the mind.				Conceptual thing	Thing, in the mind	
60-05	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).				Concrete thing		
60-05						Tangible concrete thing		
60-05						Intangible concrete thing		
60-06	The Form of a Thing is the collection of its Features.	The Form of a Concrete thing is the collection of its Features.				Concrete thing	with assigned Form	
60-08	In applying a type, the mind puts, so to speak, the type's template over the things in the world, and sieves out the ones that conform.	In applying a type, the mind puts, so to speak, the type's template over the Things in the World, and sieves out the ones that conform.				Thing	can be member of a World	cyclic
60-08						World	collection of Things	
60-08						Thing	can conform to a Type	
60-10	Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing] (also represented in Fig. 5.6 by a picture of Charlie Chaplin).	Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing] (also represented in Fig. 5.6 by a picture of Charlie Chaplin).	-	[some thing] is a placeholder, not an example		Concrete thing	with Concrete object	cyclic

	60-11	To illustrate that Things are Objects with assigned Features, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions, where every protrusion stands for a Feature.	To illustrate that Things are Objects with assigned Form, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions, where every protrusion stands for a Feature.				Thing	with Object, with assigned Form	
	60-13	When putting the template of the type human over the thing, we conclude that it conforms to the type human, because one of its protrusions fits the protrusion of the type human.	When putting the template of the Type human over the Thing, we conclude that it conforms to the Type human, because one of its protrusions fits the protrusion of the Type human.				Thing	can conform to a Type	
	60-20	The Conceptual things of facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.	The Conceptual things of Facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.				Conceptual class	type extension, collection of Conceptual things	
	60-21	Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN.	Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN.	2	is Conceptual class a collection of Things or a collection of Conceptual things?	Either (A), it is a collection of Things, or (B) it is a collection of Conceptual things	We chose (B); the previous sentence states Conceptual things			
Cycle 17	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things in reality), called referents in the semiotic triangle				Cycle 6, 14	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	in reality	
	54-02							Concrete thing	can be represented by a Conceptual thing	cyclic
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.				Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.				Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.				Cycle 7	World	collection of Things	
	55-12						Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.				Cycle 8	World	collection of Things	
	56-01							Thing	distinguishable	
	56-01							Thing	can be member of a World	cyclic
	56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.					Thing	with Object, with assigned Form	
	56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.					Thing	with Object, with assigned Form	
	56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.					[a cloud]	with Object, with assigned Form, Thing	
	56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.					Thing	with Object, with assigned Form	
	56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.					[a body]	with Object, with assigned Form	
	56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object.					Thing	with Object, with assigned Form	
		Bunge names an Object a bare individual.	Bunge names an Object a bare individual.							
	56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.					Thing	with Object	cyclic
	56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.					Thing	with Object, with assigned Form	
	56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.				Object formal context	Concrete thing	with Concrete object	cyclic
	56-19							[a human]	Concrete thing, with Concrete object	
	56-19							[a dog]	Concrete thing, with Concrete object	
	56-19							[a membership]	Concrete thing, with Concrete object	
	56-19							[a rental]	Concrete thing, with Concrete object	
	56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.					Abstract thing	with Abstract object	cyclic
	56-20							<a number>	Abstract thing, with Abstract object	
	56-20							<a boolean value>	Abstract thing, with Abstract object	

56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.			Cycle 8	World	collection of Things	
56-21						Conceptual thing	can represent a Concrete thing	
56-21						Concrete thing	can be represented by a Conceptual thing	cyclic
56-21					Object formal context	Concrete thing	with Concrete object	cyclic
56-21					Object formal context	Conceptual thing	with Conceptual object	cyclic
56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.			Cycle 9	Class	collection of Things	
56-24					Cycle 9	Thing	can be member of a Class	cyclic
56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.			Cycle 9	Class	collection of Things	
56-27					Cycle 9	Thing	can be member of a Class	cyclic
56-27						[HUMAN]	Class, collection of Things	
56-27						[RENTAL]	Class, collection of Things	
56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.				Conceptual thing	with Conceptual object	cyclic
56-28						Conceptual thing	with Object, with assigned Form	
56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.			Cycle 10	Tangible concrete thing	physical dimensions	disjoint & complete: Tangible concrete thing, Intangible concrete thing
56-29						Intangible concrete thing	conceived	
56-29						Intangible concrete thing	can be member of a conceived world	cyclic
56-29						Conceived world	collection of Intangible things	
56-29						[a human]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a dog]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a membership]	Concrete thing, Intangible concrete thing, conceived	
56-29						[a rental]	Concrete thing, Intangible concrete thing, conceived	
56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.				Concrete thing	can conform to a Type	
56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).						
56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	-	The bracket convention is not consistent	We follow the convention esthablished in the Object formal context	[the human 'Charlie Chaplin']	Concrete thing	
56-33						(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and Classes, we use the bracket pair "[" and "]".						
57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to Classes.						
57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.				[HUMAN]	Concrete class, collection of Concrete things	
57-02						Concrete class	Collection of Concrete things	
57-03	It is represented in our mind by the thought (the class HUMAN).	The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN)				(HUMAN)	Conceptual class, collectiong of Conceptual things	
57-04	The string "HUMAN" is a name from the name class for classes.	The string "HUMAN" is a name from the name class for classes.						
57-06	The referents we consider now are thus the Concrete objects of Concrete things.	The referents we consider now are thus the Concrete objects of Concrete things.						
57-09	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.						
58-02	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.				Abstract thing	with Abstract object	cyclic

58-03	As indicated by the question mark, Abstract things don't represent any Concrete things	As indicated by the question mark, Abstract things don't represent any Concrete things				Abstract thing	can not represent a Concrete thing	cyclic
58-04	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.				Concrete thing	with Concrete object	cyclic
58-04						Abstract thing	with Abstract object	cyclic
58-04								disjoint: Concrete thing, Abstract thing
58-05	Well-known abstract things are numbers.	Well-known abstract things are numbers.				<a number>	Abstract thing	
58-06	The name "the number '387'" signifies the abstract thing (the number '387').	The name "the number '387'" signifies the abstract thing (the number '387').				<a number>	Abstract thing	
58-07	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.				<a number>	Abstract thing	
58-08	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.				Thing	distinguishable	
58-14	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.				(the human 'Charlie Chaplin')	Conceptual thing	
59-01	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	If the Thing of a Unary fact is a Concrete thing, the Concrete unary fact is commonly called an Entity.			cycle 12	Thing	variable of Unary fact	cyclic
59-01						Concrete thing	variable of Entity, Thing	cyclic
59-02	As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".	As an example, one commonly speaks of the Entity Charlie Chaplin, thereby referring to the Concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".				[a human]	Concrete thing	
59-03	Likewise, if the thing of a unary fact is Abstract, the Abstract thing is commonly called a Value.	Likewise, if the Thing of a Unary fact is Abstract, the Abstract unary fact is commonly called a Value.			cycle 12	Thing	variable of Unary fact	cyclic
59-03						Abstract thing	variable of Value, Thing	cyclic
59-04	As an example, day 2411108 is a value on the time scale Julian Date.	As an example, date 2411108 is a value on the time scale Julian Date.			cycle 13	<a date>	Abstract thing, Thing	
59-07	In answering these questions, we will adhere strictly to the distinctions we have made between concrete and conceptual things.	In answering these questions, we will adhere strictly to the distinctions we have made between Concrete and Conceptual things.				Concrete thing		disjoint: Concrete thing, Conceptual thing
59-07						Conceptual thing		
59-08	Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.	Concrete things are the Things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.			Cycle 15	Concrete thing	Thing, in reality	disjoint & complete: inside a mind, in reality
59-08						[an insurance policy]	Concrete thing, in reality, Thing	
59-08						[a human]	Concrete thing, in reality, Thing	
59-08						[a car]	Concrete thing, in reality, Thing	
59-08						[a house]	Concrete thing, in reality, Thing	
59-08						[a rental]	Concrete thing, in reality, Thing	
59-08						[an employment contract]	Concrete thing, in reality, Thing	
59-09	As said, concrete things can be tangible (like a house) and intangible (like a rental).	As said, concrete things can be tangible (like a house) and intangible (like a rental).				Tangible concrete thing	Concrete thing	disjoint: Tangible concrete thing, Intangible concrete thing
59-09						Intangible concrete thing	Concrete thing	
59-09						[a house]	Concrete thing, Tangible concrete thing	
59-09						[a rental]	Concrete thing, Intangible concrete thing	
59-10	Conceptual things are things in the mind.	Conceptual things are Things in the mind.				Conceptual thing	Thing, in a mind	
60-05	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).				Concrete thing		
60-05						Tangible concrete thing		
60-05						Intangible concrete thing		
60-06	The Form of a Thing is the collection of its Features.	The Form of a Concrete thing is the collection of its Features.				Concrete thing	with assigned Form	
60-08	In applying a type, the mind puts, so to speak, the type's template over the things in the world, and sieves out the ones that conform.	In applying a type, the mind puts, so to speak, the type's template over the Things in the World, and sieves out the ones that conform.				Thing	can be member of a World	cyclic
60-08						World	collection of Things	
60-08						Thing	can conform to a Type	
60-10	Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing] (also represented in Fig. 5.6 by a picture of Charlie Chaplin).	Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing] (also represented in Fig. 5.6 by a picture of Charlie Chaplin).	-	[some thing] is a placeholder, not an example		Concrete thing	with Concrete object	cyclic

	60-11	To illustrate that Things are Objects with assigned Features, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions, where every protrusion stands for a Feature.	To illustrate that Things are Objects with assigned Form, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions, where every protrusion stands for a Feature.					Thing	with Object, with assigned Form	
	60-13	When putting the template of the type human over the thing, we conclude that it conforms to the type human, because one of its protrusions fits the protrusion of the type human.	When putting the template of the Type human over the Thing, we conclude that it conforms to the Type human, because one of its protrusions fits the protrusion of the Type human.					Thing	can conform to a Type	
	60-20	The Conceptual things of facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.	The Conceptual things of Facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.					Conceptual class	type extension, collection of Conceptual things	
	60-21	Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN.	Let us call the Conceptual class of Conceptual things that are predicated by Facts of the Type human, (HUMAN).				Cycle 15	Conceptual class	type extension, collection of Conceptual things	
	60-21							(HUMAN)	type extension, collection of Conceptual things, Conceptual class	
	60-22	This class is called the extension of the type human.	This Conceptual class is called the extension of the Type human.	-	Challenge for analyse of Type concept: are these two terms synonyms?		Cycle 15	Conceptual class	type extension	
	60-22						Cycle 15	(HUMAN)	type extension, Conceptual class	
	60-23	Conversely, the type human is called the intension of the class HUMAN.	Conversely, the type human is called the intension of the class HUMAN.	-	Challenge for analyse of Type concept: are these two terms synonyms?			Conceptual class	with type intension	
	60-23							(HUMAN)	with type intension, Conceptual class	
	60-25	Lastly, the conceptual class (HUMAN) represents the concrete class [HUMAN].	Lastly, the Conceptual things of the Conceptual class (HUMAN) represents the Concrete things of Concrete class [HUMAN].				Cycle 11	Conceptual class	collection of Conceptual things	
	60-25							Concrete class	collection of Concrete things	
	60-25							Conceptual thing	can represent a Concrete thing	
	60-25							Concrete thing	can be represented by a Conceptual thing	cyclic
	60-25							(HUMAN)	Conceptual class, collection of Conceptual things	
	60-25							[HUMAN]	Concrete class, collection of Concrete things	
	61-04	A Fact starts to exist either as soon as the Form of a thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms.	A Fact starts to exist either as soon as the Form of a Thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms.		2 Does a Thing conform to a Type, or does its Form conform to a Type? Both are possible in this sentence	Either (A) a Thing conforms to a Type, or (B) its Form conforms to a Type	We chose (A). Bunge confirms that a Form without an Object cannot exist; confirming the statements that a Thing may conform to a Type.			
	Key	**Sentence**	**Interpretation**	**Category**	**Issue**	**Options**	**Resolution**	**Example/concept sign**	**Characteristics**	
Cycle 18	54-01	Thoughts are things in the mind.	Some thoughts are Conceptual things in the mind.				Cycle 5, 6	Conceptual thing	in a mind	
	54-02	They represent concrete things (i.e. things outside the mind), called referents.	They represent Concrete things (i.e. Things in reality), called referents in the semiotic triangle				Cycle 6, 14	Conceptual thing	can represent a Concrete thing	disjoint: Concrete thing, Conceptual thing
	54-02							Concrete thing	in reality	
	54-02							Concrete thing	can be represented by a Conceptual thing	cyclic
	54-09	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.	Lastly, although the relationship between a sign and its referent is completely determined by their being connected through the thought, it is often also indicated separately: the sign is said to denote the referent.				Cycle 5			
	54-11	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.	The semiotic triangle is a quite useful but also a quite simplified representation of the three core concepts in communication: sign, thought, and referent, and the relationships between them.				Cycle 5			
	55-12	Bunge defines a world as a collection of things, 'untouched' yet by mental abstractions.	Bunge defines a world as a collection of Things, 'untouched' yet by mental abstractions.				Cycle 7	World	collection of Things	
	55-12						Cycle 7	Thing	can be member of a World	cyclic
	56-01	The basic assumption in world ontology is that a world consists of distinguishable concrete things or things for short.	The basic assumption in world ontology is that a World consists of distinguishable Things.				Cycle 8	World	collection of Things	
	56-01							Thing	distinguishable	
	56-01							Thing	can be member of a World	cyclic
	56-02	A Thing is said to be an Object with assigned Features.	A Thing is said to be an Object with an assigned Form of Features.					Thing	with Object, with assigned Form	
	56-03	Only through these Features can the Thing be recognised; an Object without features is no thing.	Only through the Features of its Form can the Thing be recognised; an Object without Form is no thing.					Thing	with Object, with assigned Form	
	56-05	The cloud is a Thing, of which the main Features are the recognisable water drops or crystals.	The cloud is a Thing, of which the main Features are the recognisable water drops and crystals.					[a cloud]	with Object, with assigned Form, Thing	
	56-09	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Thing has no (more) recognisable Features.	In many cases, however, it makes sense to conceive an Object, i.e. a persisting identity, of a Thing, even if the Form of a Thing has no (more) recognisable Features.					Thing	with Object, with assigned Form	
	56-12	Although the Features of a body do change dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.	Although the Form of a body changes dramatically during its lifetime, as we know, we are inclined to say that it is still the same body.					[a body]	with Object, with assigned Form	

56-13	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that the Features of a Thing are assigned to an Object. Bunge names an Object a bare individual.	The key to Bunge's solution of the problem of sameness and change, which we will adopt in Sect. 5.3.2, is that a Thing is a Form, a collection of Features, that is assigned to an Object. Bunge names an Object a bare individual.				Thing	with Object, with assigned Form	
56-14	In this bare Object resides the identity of the thing.	In this bare Object resides the identity of the thing.				Thing	with Object	cyclic
56-16	Only when at least one Feature is recognised, there is a thing.	Only when a Form has at least one Feature, there is a Thing.				Thing	with Object, with assigned Form	
56-19	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.	Concrete objects are Objects of Concrete things, like humans, dogs, memberships and rentals.			Object formal context	Concrete thing	with Concrete object	cyclic
56-19						[a human]	Concrete thing, with Concrete object	
56-19						[a dog]	Concrete thing, with Concrete object	
56-19						[a membership]	Concrete thing, with Concrete object	
56-19						[a rental]	Concrete thing, with Concrete object	
56-20	Abstract objects are Objects of Abstract things, like numbers and boolean values.	Abstract objects are Objects of Abstract things, like numbers and boolean values.				Abstract thing	with Abstract object	cyclic
56-20						<a number>	Abstract thing, with Abstract object	
56-20						<a boolean value>	Abstract thing, with Abstract object	
56-21	When observing and distinguishing the Concrete things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.	When recognising and distinguishing the Things in a world, the mind creates Conceptual things that represent the Concrete Things. Conceptual objects are Objects of Conceptual things. This implicates that the Conceptual object of a Conceptual thing represents the Concrete object of a Concrete thing.			Cycle 8	World	collection of Things	
56-21						Conceptual thing	can represent a Concrete thing	
56-21						Concrete thing	can be represented by a Conceptual thing	cyclic
56-21					Object formal context	Concrete thing	with Concrete object	cyclic
56-21					Object formal context	Conceptual thing	with Conceptual object	cyclic
56-24	It is called classifying: we observe similarities and differences between things and accordingly put them into different classes.	It is called classifying: we recognise similarities and differences between Things and accordingly put them into different Classes.			Cycle 9	Class	collection of Things	
56-24					Cycle 9	Thing	can be member of a Class	cyclic
56-27	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we observe, and then conclude that the Things belong to certain Classes, like humans and rentals.	As will be discussed in Sect. 5.2.4, Types serve as templates that we 'put on' the Things we recognise, and then conclude that the Things belong to certain Classes, like humans and rentals.			Cycle 9	Class	collection of Things	
56-27					Cycle 9	Thing	can be member of a Class	cyclic
56-27						[HUMAN]	Class, collection of Things	
56-27						[RENTAL]	Class, collection of Things	
56-28	Consequently, Conceptual things are Conceptual objects with assigned Features	Consequently, Conceptual things are Conceptual objects with assigned Form, a collection of Features.				Conceptual thing	with Conceptual object	cyclic
56-28						Conceptual thing	with Object, with assigned Form	
56-29	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a conceived world, like memberships and rentals.	Fig. 5.3 (left) illustrates the foregoing by applying the semiotic triangle to Concrete things, either tangible, so things that one can observe, like humans and dogs, or intangible, i.e. things that one considers to exist in a Conceived world, like memberships and rentals.			Cycle 10	Tangible concrete thing	physical dimensions	disjoint & complete: Tangible concrete thing, Intangible concrete thing
56-29						Intangible concrete thing	conceived	
56-29						Intangible concrete thing	can be member of a conceived world	cyclic
56-29						Conceived world	collection of Intangible things	
56-29						[a human]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a dog]	Concrete thing, Tangible concrete thing, physical dimensions	
56-29						[a membership]	Concrete thing, Intangible concrete thing, conceived	
56-29						[a rental]	Concrete thing, Intangible concrete thing, conceived	
56-30	The example referent in Fig. 5.3 (left) is a concrete thing of the type human.	The example referent in Fig. 5.3 (left) is a Concrete thing of the Type human.				Concrete thing	can conform to a Type	
56-32	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).	The string "Charlie Chaplin" belongs to a specific name class that is used to refer to humans (but it may also belong to other name classes, of course).						
56-33	To avoid confusion, we will refer to the concrete thing as <the human 'Charlie Chaplin'>, and to the corresponding conceptual thing as (the human 'Charlie Chaplin').	To avoid confusion, we will refer to the Concrete thing as [the human 'Charlie Chaplin'], and to the corresponding Conceptual thing as (the human 'Charlie Chaplin').	-	The bracket convention is not consistent	We follow the convention esthablished in the Object formal context	[the human 'Charlie Chaplin']	Concrete thing	
56-33						(the human 'Charlie Chaplin')	Conceptual thing, can represent a Concrete thing	
56-34	For Concrete objects and classes, we use the bracket pair "[" and "]".	For Concrete objects and Classes, we use the bracket pair "[" and "]".						

57-01	Fig. 5.3 (right) shows an example of applying the semiotic triangle to classes.	Fig. 5.3 (right) shows an example of applying the semiotic triangle to Classes.				
57-02	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.	The name "the class HUMAN" denotes the Concrete class [HUMAN], which is the set of the Concrete things that are humans.		[HUMAN]	Concrete class, collection of Concrete things	
57-02				Concrete class	Collection of Concrete things	
57-03	It is represented in our mind by the thought (the class HUMAN).	The Concrete things of Class [HUMAN] are represented by Conceptual things from Class (HUMAN)		(HUMAN)	Conceptual class, collectiong of Conceptual things	
57-04	The string "HUMAN" is a name from the name class for classes.	The string "HUMAN" is a name from the name class for classes.				
57-06	The referents we consider now are thus the Concrete objects of Concrete things.	The referents we consider now are thus the Concrete objects of Concrete things.				
57-09	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.	The numeric code "701" is arbitrarily chosen from the name class that is used to uniquely signify Conceptual things.				
58-02	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.	Fig. 5.4 (right) shows an example of applying the semiotic triangle to Abstract things that have Abstract objects.		Abstract thing	with Abstract object	cyclic
58-03	As indicated by the question mark, Abstract things don't represent any Concrete things	As indicated by the question mark, Abstract things don't represent any Concrete things		Abstract thing	can not represent a Concrete thing	cyclic
58-04	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.	Through reasoning by analogy (with Concrete objects and Concrete things), however, we will conceive Abstract objects and Abstract things.		Concrete thing	with Concrete object	cyclic
58-04				Abstract thing	with Abstract object	cyclic
58-04						disjoint: Concrete thing, Abstract thing
58-05	Well-known abstract things are numbers.	Well-known abstract things are numbers.		<a number>	Abstract thing	
58-06	The name "the number '387'" signifies the abstract thing (the number '387').	The name "the number '387'" signifies the abstract thing (the number '387').		<a number>	Abstract thing	
58-07	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.	As examples of using this number, we may have counted 387 humans or dogs or memberships or rentals.		<a number>	Abstract thing	
58-08	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.	Let us now generalise the Features that Things may have and through which we are able to recognise and distinguish them, in the notion of Fact.		Thing	distinguishable	
58-14	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.	The example conceptual fact is that the Conceptual thing signified by (object 701) is a human.		(the human 'Charlie Chaplin')	Conceptual thing	
59-01	If the thing of a Unary fact is a Concrete thing, the Concrete thing is commonly called an Entity.	If the Thing of a Unary fact is a Concrete thing, the Concrete unary fact is commonly called an Entity.	cycle 12	Thing	variable of Unary fact	cyclic
59-01				Concrete thing	variable of Entity, Thing	cyclic
59-02	As an example, one commonly speaks of the entity Charlie Chaplin, thereby referring to the concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".	As an example, one commonly speaks of the Entity Charlie Chaplin, thereby referring to the Concrete unary fact that object 701 is a human, denoted by the name "Charlie Chaplin".		[a human]	Concrete thing	
59-03	Likewise, if the thing of a unary fact is Abstract, the Abstract thing is commonly called a Value.	Likewise, if the Thing of a Unary fact is Abstract, the Abstract unary fact is commonly called a Value.	cycle 12	Thing	variable of Unary fact	cyclic
59-03				Abstract thing	variable of Value, Thing	cyclic
59-04	As an example, day 2411108 is a value on the time scale Julian Date.	As an example, date 2411108 is a value on the time scale Julian Date.	cycle 13	<a date>	Abstract thing, Thing	
59-07	In answering these questions, we will adhere strictly to the distinctions we have made between concrete and conceptual things.	In answering these questions, we will adhere strictly to the distinctions we have made between Concrete and Conceptual things.		Concrete thing		disjoint: Concrete thing, Conceptual thing
59-07				Conceptual thing		
59-08	Concrete things are the things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.	Concrete things are the Things that we deal with in reality: the people we cooperate with, the car we drive in, the house we live in, but also the rental contract of our house, the insurance policy for our car, and the employment contract we have with our employer.	Cycle 15	Concrete thing	Thing, in reality	disjoint & complete: inside a mind, in reality
59-08				[a human]	Concrete thing, in reality, Thing	
59-08				[a car]	Concrete thing, in reality, Thing	
59-08				[a house]	Concrete thing, in reality, Thing	
59-08				[a rental]	Concrete thing, in reality, Thing	
59-08				[an insurance policy]	Concrete thing, in reality, Thing	
59-08				[an employment contract]	Concrete thing, in reality, Thing	
59-09	As said, concrete things can be tangible (like a house) and intangible (like a rental).	As said, concrete things can be tangible (like a house) and intangible (like a rental).		Tangible concrete thing	Concrete thing	disjoint: Tangible concrete thing, Intangible concrete thing
59-09				Intangible concrete thing	Concrete thing	
59-09				[a house]	Concrete thing, Tangible concrete thing	
59-09				[a rental]	Concrete thing, Intangible concrete thing	
59-10	Conceptual things are things in the mind.	Conceptual things are Things in the mind.		Conceptual thing	Thing, in a mind	
60-05	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).	A Type is defined as a template that the mind applies when we consider Concrete things (both tangible and intangible).		Concrete thing		
60-05				Tangible concrete thing		

60-05						Intangible concrete thing		
60-06	The Form of a Thing is the collection of its Features.	The Form of a Concrete thing is the collection of its Features.				Concrete thing	with assigned Form	
60-08	In applying a type, the mind puts, so to speak, the type's template over the things in the world, and sieves out the ones that conform.	In applying a type, the mind puts, so to speak, the type's template over the Things in the World, and sieves out the ones that conform.			Cycle 15	Thing	can be member of a World	cyclic
60-08						World	collection of Things	
60-08						Thing	can conform to a Type	
60-10	Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing] (also represented in Fig. 5.6 by a picture of Charlie Chaplin).	Let us next assume that there is a Concrete thing whose Concrete object we signify by [object 701], and let us study the Concrete thing that is indicated as [some thing] (also represented in Fig. 5.6 by a picture of Charlie Chaplin).	-	[some thing] is a placeholder, not an example		Concrete thing	with Concrete object	cyclic
60-11	To illustrate that Things are Objects with assigned Features, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions, where every protrusion stands for a Feature.	To illustrate that Things are Objects with assigned Form, the symbol for the Thing in the semiotic mill is a magenta coloured disk with protrusions, where every protrusion stands for a Feature.				Thing	with Object, with assigned Form	
60-13	When putting the template of the type human over the thing, we conclude that it conforms to the type human, because one of its protrusions fits the protrusion of the type human.	When putting the template of the Type human over the Thing, we conclude that it conforms to the Type human, because one of its protrusions fits the protrusion of the Type human.				Thing	can conform to a Type	
60-20	The Conceptual things of facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.	The Conceptual things of Facts of the same type, constitute a Conceptual class of which each of the Conceptual things is a member.				Conceptual class	type extension, collection of Conceptual things	
60-21	Let us call the Conceptual class of Things that are predicated by Facts of the Type human, HUMAN.	Let us call the Conceptual class of Conceptual things that are predicated by Facts of the Type human, (HUMAN).			Cycle 13	Conceptual class	type extension, collection of Conceptual things	
60-21						(HUMAN)	type extension, collection of Conceptual things, Conceptual class	
60-22	This class is called the extension of the type human.	This Conceptual class is called the extension of the Type human.	-	Challenge for analyse of Type concept: are these two terms synonyms?	Cycle 16	Conceptual class	type extension	
60-22					Cycle 16	(HUMAN)	type extension, Conceptual class	
60-23	Conversely, the type human is called the intension of the class HUMAN.	Conversely, the type human is called the intension of the class HUMAN.	-	Challenge for analyse of Type concept: are these two terms synonyms?		Conceptual class	with type intension	
60-23						(HUMAN)	with type intension, Conceptual class	
60-25	Lastly, the conceptual class (HUMAN) represents the concrete class [HUMAN].	Lastly, the Conceptual things of the Conceptual class (HUMAN) represents the Concrete things of Concrete class [HUMAN].			Cycle 16	Conceptual class	collection of Conceptual things	
60-25						Concrete class	collection of Concrete things	
60-25						Conceptual thing	can represent a Concrete thing	
60-25						Concrete thing	can be represented by a Conceptual thing	cyclic
60-25						(HUMAN)	Conceptual class, collection of Conceptual things	
60-25						[HUMAN]	Concrete class, collection of Concrete things	
61-04	A Fact starts to exist either as soon as the Form of a thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms.	A Fact starts to exist either as soon as the Thing conforms to a Type, or as soon as a Type starts to exist to which the Thing conforms.			Cycle 17	Thing	can conform to a Type	
61-06	Through this event, a new entity of the type butterfly starts to exist.	Through this event, a new Entity of the Type butterfly starts to exist.						
61-09	A Fact ends to exist either as soon as the Form of the Thing that the Fact is about, does not conform anymore to the Type, or as soon as a Type to which it conforms, ends to exist.	A Fact ends to exist either as soon as the Thing that the Fact is about, does not conform anymore to the Type, or as soon as a Type to which it conforms, ends to exist.				Thing	can conform to a Type	
61-11	After this event has occurred, the thing does not conform anymore to the type caterpillar.	After this event has occurred, the Thing does not conform anymore to the Type caterpillar.				Thing	can conform to a Type	
61-12	Consequently, the entity of the type caterpillar ends to exist.	Consequently, the Entity of the Type caterpillar ends to exist.						
61-16	As Mario Bunge puts it, a world is not a collection of facts, pace Wittgenstein, but a collection of things [5].	As Mario Bunge puts it, a World is not a collection of Facts, pace Wittgenstein, but a collection of Things [5].				World	collection of Things	
61-16						Thing	can be member of a World	cyclic
61-17	The knowledge of these things, and thus one's factual knowledge of the world, consists of conceptual facts [9].	The knowledge of these Things, and thus one's factual knowledge of the world, consists of Conceptual facts [9].				Conceptual fact		
62-09	As is explained in Chap. 8, the entity types that occur in product kinds are proper candidates for being declared.	As is explained in Chap. 8, the Entity types that occur in Product kinds are proper candidates for being declared.						
62-21	When this is not the case anymore, the student entity ceases to exist.	When this is not the case anymore, the student Entity ceases to exist.				[a student]	Concrete thing	
62-22	By generalisation is understood that one defines a new type as the union of the extensions, so the classes, of two or more existing types (which may be declared or derived).	By generalisation is understood that one defines a new Type as the union of the extensions, so the Classes, of two or more existing types (which may be declared or derived).				Class	type extension	
62-24	We can then define the type vehicle as the intension of the class (cf. Fig. 5.7) that equals the set-theoretic union of the extensions of car, boat and aircraft.	We can then define the Type vehicle as the intension of the Class (cf. Fig. 5.7) that equals the set-theoretic union of the extensions of car, boat and aircraft.				Class	with type intension, type extension	

62-24			-		We use [CAR], based on 57-02	[CAR]	with type intension, type extension
62-24			-		We use [BOAT], based on 57-02	[BOAT]	with type intension, type extension
62-24			-		We use [AIRCRAFT], based on 57-02	[AIRCRAFT]	with type intension, type extension
64-01	The individuals in all these classes are identified by the concrete identity they have got in the class genus.	The Things in all these Classes are identified by the Things in the class genus.				Class	collection of Things
64-11	The type human, as discussed, is typically a constructional type: it is based exclusively on the properties of things.	The Type human, as discussed, is typically a constructional Type: it is based exclusively on the Form of a Thing			Feature formal context		
65-01	Figure 5.9 exhibits the conceptualisation of things of the type sit-on-able.	Figure 5.9 exhibits the conceptualisation of Things of the Type sit-on-able.				Thing	can conform to a Type
65-02	It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory.	It shows the sit-on-ability relationship between the Thing (Charlie Chaplin) and the Thing (this chair). Relationships are detailed in the TAO theory.	-	Using same symbols as 56-33 for Charlie Chaplin		(the human 'Charlie Chaplin') (this chair)	Thing Thing
65-11	The type 'chair model Chaplin' can be specified precisely enough from the construction perspective, in order to decide whether a thing conforms to the type or not.	The Type 'chair model Chaplin' can be specified precisely enough from the construction perspective, in order to decide whether a Thing conforms to the Type or not.				Thing	can conform to a Type
66-03	Its general questions are: When does a thing change? and: If a thing changes, is it still the same thing (only changed)?	Its general questions are: When does a Thing change? and: If a Thing changes, is it still the same Thing (only changed)?					
66-05	The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place.	The ship wherein Theseus and the youth of Athens returned from a long journey was preserved by the Athenians by taking away the old planks as they decayed, putting in new and stronger timber in their place.				[a ship] [a plank]	Composite thing, collection of Things Thing
66-06	In the end, all parts of Theseus' ship are replaced.	In the end, all parts of Theseus' ship are replaced.				[a ship] [a plank]	Composite thing, collection of Things Thing
66-08	The key to an appropriate solution of the problem of sameness and change, exemplified by the paradox of Theseus, is to recognise that every Thing has an Object, that is taken as its identity, next to having an assigned Form, a collection of Features.	The key to an appropriate solution of the problem of sameness and change, exemplified by the paradox of Theseus, is to recognise that every Thing has an Object, that is taken as its identity, next to having an assigned Form, a collection of Features.				Thing	with Object, with assigned Form
66-09	So, even if all the parts of the ship are replaced by other (similar) ones, it is still the same ship.	So, even if all the parts of the ship are replaced by other (similar) ones, it is still the same ship.				[a ship]	Thing, with Object, with assigned Form
66-10	This argument is supported by the distinction between matter-constant (the ship is composed of the same part instances) and form-constant (the ship is an instance of the type ship), as proposed by Simons [10].	This argument is supported by the distinction between matter-constant (the ship is composed of the same parts) and form-constant (the shipconforms to the Type ship), as proposed by Simons [10].				[a ship]	Composite thing, collection of Things
66-11	When the first plank is replaced, the ship is already not the same matter-constant ship anymore.	When the first plank is replaced, the ship is already not the same matter-constant ship anymore.				[a plank]	Thing
66-11	When the first plank is replaced, the ship is already not the same matter-constant ship anymore.	When the first plank is replaced, the ship is already not the same matter-constant ship anymore.				[a ship]	Composite thing, collection of Things
66-12	The form-constant argument preserves that it is still a ship of the same type because the whole thing keeps conforming to the type ship (even when all parts are replaced).	The form-constant argument preserves that it is still a ship of the same Type because the whole Thing keeps conforming to the Type ship (even when all parts are replaced).				[a ship]	Composite thing, collection of Things, can conform to a Type
66-13	However, our argument goes further: Theseus' ship does indeed still conform to the Type ship, but, in addition, it has kept its Object.	However, our argument goes further: Theseus' ship does indeed still conform to the Type ship, but, in addition, it has kept its Object.				[a ship]	can conform to a Type, with Object
66-14	This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before.	This would even hold if the Thing would not conform to the Type ship anymore, as a butterfly has kept its Object after having transformed to being a butterfly from being a caterpillar before.				[a ship]	Thing, with Object
66-14						[a butterfly]	Thing, with Object
66-14						[a caterpillar]	Thing, with Object
66-16	Assume that all parts are replaced and that the 'old' ship is rebuilt by assembling the 'old' parts.	Assume that all parts are replaced and that the 'old' ship is rebuilt by assembling the 'old' parts.				[a ship]	Composite thing, collection of Things
66-17	Then which ship is Theseus' ship?	Then which ship is Theseus' ship?				[a ship]	Thing
66-19	We are now confronted with two concrete ships, each of them having its own Concrete object.	We are now confronted with two concrete ships, each of them having its own Concrete object.				[a ship]	Concrete thing, with Concrete object
66-20	Most likely, Theseus would consider the current one his ship, not the one that has been re-assembled from the thrown away parts.	Most likely, Theseus would consider the current one his ship, not the one that has been re-assembled from the thrown away parts.				[a ship]	Composite thing, collection of Things
66-24	Many researchers in ontology, like Guarino [11] and Guizzardi [12], try to solve the problem of sameness and change by assigning the identity of a composite thing to one of its components.	Many researchers in ontology, like Guarino [11] and Guizzardi [12], try to solve the problem of sameness and change by assigning the identity of a Composite thing to one of its components.				Composite thing	collection of Things

66-25	Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore.	Regarding the human body, they consider the human brains as the component that bears the body's identity: as soon as the brains are replaced, it is not the same body anymore.		[a body]	Composite thing, collection of Things
66-25				[a brain]	Thing
66-26	With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity.	With respect to cars, they choose a particular part (mostly the body of the car or the motor block) to bear the car's identity.		[a car]	Composite thing, collection of Things
66-26				[a motor block]	Thing
67-19	The important point to recognise is that a Composite thing has its own Object, independent of the Objects of its components.	The important point to recognise is that a Composite thing has its own Object, independent of the Objects of its components.		Composite thing	collection of Things, with Object
68-09	The same 'business entity' (upper part) appears to conform to two types, namely to the type purchase (left side), and to the type sale (right side), respectively applied by the buyer and the seller enterprise.	The same 'business entity' (upper part) appears to conform to two Types, namely to the Type purchase (left side), and to the Type sale (right side), respectively applied by the buyer and the seller enterprise.			
68-12	Because the notions of purchase and sale apply to the same business thing, they are called dual notions.	Because the notions of purchase and sale apply to the same Thing, they are called dual notions.		Thing	can conform to a Type
69-03	Two names are called synonymous if they signify the same thought (or denote the same referent, cf. Fig. 5).	Two names are called synonymous if they signify the same thought (or denote the same referent, cf. Fig. 5).			
69-06	Instead they refer to different things, with different properties.	Instead they refer to different Things, with different Features.	Feature formal context		

disjoint & complete: inside a mind, in reality